

Problem C. Teleport Maze

Time Limit 2000 ms

Problem Statement

There is a maze consisting of a grid with H rows and W columns. Let (i, j) denote the cell at the i -th row from the top and j -th column from the left. The type of cell (i, j) is given as a character $S_{i,j}$, where each character has the following meaning:

- . : Empty cell
- # : Obstacle cell
- Lowercase English letter (a - z): Warp cell

In the maze, you can perform the following two types of actions any number of times in any order:

- **Walk:** Move from the current cell to a cell that is one cell away in one of the four directions (up, down, left, right). However, you cannot move to an obstacle cell or outside the grid.
- **Warp:** When you are at a warp cell, move to any warp cell with the same character written on it.

Determine whether it is possible to move from cell $(1, 1)$ to cell (H, W) , and if possible, find the minimum total number of actions required.

Constraints

- $1 \leq H, W \leq 1000$
- $H \times W \geq 2$
- H and W are integers.
- $S_{i,j}$ is ., #, or a lowercase English letter.
- $S_{1,1} \neq \text{#}$
- $S_{H,W} \neq \text{#}$

Input

The input is given from Standard Input in the following format:

$$\begin{array}{c} H \ W \\ S_{1,1} S_{1,2} \dots S_{1,W} \\ \vdots \\ S_{H,1} S_{H,2} \dots S_{H,W} \end{array}$$

Output

If it is possible to move from cell $(1, 1)$ to cell (H, W) , print the minimum total number of actions required; otherwise, print **-1**.

Sample 1

Input	Output
<pre>3 4 ..a. #### ba#b</pre>	5

You can move from cell $(1, 1)$ to cell $(3, 4)$ by performing actions as follows:

1. Move from cell $(1, 1)$ to cell $(1, 2)$ by walking.
2. Move from cell $(1, 2)$ to cell $(1, 3)$ by walking.
3. Move from cell $(1, 3)$ to cell $(3, 2)$ by warping.
4. Move from cell $(3, 2)$ to cell $(3, 1)$ by walking.
5. Move from cell $(3, 1)$ to cell $(3, 4)$ by warping.

The total number of actions is 5, which is the minimum.

Sample 2

Input	Output
<pre>3 4 ..a. #### b.#b</pre>	-1

It is impossible to move from cell $(1, 1)$ to cell $(3, 4)$.

Sample 3

Input	Output
4 4 XXXX XXXX XXXX XXXX	1

Sample 4

Input	Output
7 11 u..#y..#... k..#.z.#.k. iju#...#x.. ##### ..x#.t.#..n abc#y..#... ..z#..t#.y.	12